Isaac J. Newton



Electronic & Computer Engineer

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PROFILE

Electronic and Computer Engineer (B.Eng., University of Nigeria, Nsukka) working across embedded systems, electronics, backend engineering and full-stack development. I build reliable software and intelligent hardware, design scalable systems, and turn ideas into functional products with real-world impact.

EXPERIENCE

Backend Engineer · IEU Africa — Remote · 2025 – Present

- Build and maintain scalable backend services and APIs for modern web applications.
- Implement authentication systems and secure access to platform features.
- Run and maintain cloud-based infrastructure.

Software and Hardware Engineer · OneGrid Energies — Remote · 2024 – Present

- Develop embedded and energy-monitoring systems.
- Build and maintain scalable web services.
- Interface hardware components with software platforms.
- Work on intelligent power and metering solutions.

Chief Technology Officer · Switch Electric — Jul 2021 – Present

- Lead technology development across embedded systems and energy solutions.
- Oversee backend infrastructure and product engineering.

Electronic Engineer · Arone Energy — 2021 – 2022

- Designed, developed and tested electronic systems and embedded hardware solutions.
- Handled circuit design and microcontroller integration.
- Troubleshot and debugged electronic systems.

PROJECTS

Foot Pressure Measurement Device

Designed and developed a foot pressure measurement device using PCB design tools. The device captures and analyses foot pressure data for applications in healthcare, sports science and gait analysis, providing insights into foot health and performance.

Tech: PCB Design, EasyEDA, Embedded Systems, Sensor Integration, Data Analysis

SoleSense PWA Insole Monitor — fluxion9.github.io/solesense/app

Built a progressive web app for SoleSense that connects to the foot pressure measurement device over Bluetooth, collects pressure data, and displays real-time monitoring and analysis for insole performance.

Tech: PWA Development, Bluetooth, Data Visualization, Sensor Data, Health Tech

OneGrid Energies · Developer

onegridenergies.com

Built the OneGrid Energies website for a Nigerian clean energy company focused on renewable energy access, solar installations, recycling innovation, and community empowerment across underserved communities.

Tech: Website Development, Frontend Development, Responsive Design, Clean Energy

Veriss Payments

payment.veriss.co

Fintech payment infrastructure for secure, seamless digital transactions. Focused on payment processing, transaction management, and scalable financial operations.

Tech: Fullstack Development, Node.js, Express.js, Vue.js, MongoDB, Paystack API

BoqBox

boqbox.ng

Worked on BoqBox, an e-commerce and gifting platform that enables users to purchase products for themselves or send them as gifts to others, with support for recurring subscription-based deliveries and seamless online shopping experiences.

Tech: Backend Development, Node.js, Express.js, Paystack API, MongoDB

Aegis pest Detection Rover · Embedded System Engineer

eneontechnologies.com/projects/aegis-pest-detection-rover

Aegis autonomously scans a farm plot, using a computer vision algorithm to identify soil and crop pests as it moves. Each detection sends a notification straight to the owner's phone on Telegram, so action can be taken before the damage spreads. Pests are often spotted too late, after the harvest is already at risk. Aegis patrols the plot consistently, replacing occasional manual inspection with regular, reliable coverage.

- Robotics
- Embedded Systems
- Computer Vision

Tech: IOT, Embedded System, ProductInnovation, Robotics and Engineering

SafeGuard LPG · Embedded System Engineer

eneontechnologies.com/projects/safeguard-lpg

SafeGuard LPG continuously monitors gas level, smoke and temperature, and triggers an alarm that corresponds to the event it detects so a gas leak, a fire risk and overheating are each clearly identified. An alarm that sounds too often is easy to ignore. SafeGuard LPG uses an adaptive learning mechanism that learns the normal conditions of its environment, reducing false triggers over time while staying responsive to real hazards.

- Hardware
- Embedded systems

Tech: Temperature, Hardware, Engineering, Software and Embedded System

Onegrid OneMeter · Software and Hardware Engineer

2025-2026

Smart Energy's metering for OneGrid charging stations: a prepaid meter that switches a load, logs device charging (oneplastic lanterns and other devices), and reports to an online dashboard.

- Supports over-the-air updates
- Uses a Touch-Screen Display
- Supports offline data logging for unstable internet
- Uses TOTP token generation for offline device activation

SKILLS

Languages C · C++ · Python · JavaScript · PHP · HTML · CSS

Backend & web	Node.js · Express.js · MongoDB · REST APIs · Paystack API · Progressive Web Apps · Responsive Design
Hardware & embedded	Embedded Systems · Hardware / Circuit Design · PCB Design · EasyEDA · Microcontrollers · Sensor Integration · Bluetooth · Internet of Things (IoT)

EDUCATION

Bachelor of Engineering (B.Eng.), Electronic and Computer Engineering — Sep 2019 – 2024

University of Nigeria, Nsukka · Nsukka, Nigeria

Portfolio, projects and the latest version of this CV: **fluxion9.github.io**